Assignment - II

Windspeed (m/s)	Blade Angle (°)	Rotor speed (RPM)	Power Output.
5	10	120	45
7	12	150	70
9	14	180	102
11	16	210	138
13	18	240	180

$\therefore * \boxed{\theta = (X^T X)^{-1} X^T Y}$

$$X = \begin{bmatrix} 1 & 5 & 10 & 120 \\ 1 & 7 & 12 & 150 \\ 1 & 9 & 14 & 180 \\ 1 & 11 & 16 & 210 \\ 1 & 13 & 18 & 240 \end{bmatrix} \quad Y = \begin{bmatrix} 45 \\ 70 \\ 102 \\ 138 \\ 180 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 10 & 12 & 14 & 16 & 18 \\ 120 & 150 & 180 & 210 & 240 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 5 & 10 & 120 \\ 1 & 7 & 12 & 150 \\ 1 & 9 & 14 & 180 \\ 1 & 11 & 16 & 210 \\ 1 & 13 & 18 & 240 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 10 & 12 & 14 & 16 & 18 \\ 120 & 150 & 180 & 210 & 240 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 45 & 70 & 900 \\ 45 & 445 & 670 & 9300 \\ 70 & 670 & 1020 & 13800 \\ 900 & 9300 & 13800 & 167400 \end{bmatrix}$$

$$X^TY = \begin{bmatrix} 535 \\ 5669 \\ 8470 \\ 105840 \end{bmatrix}$$

$$\therefore \quad \theta = (X^TX)^{-1}X^TY$$

$$\cancel{(X^TX)^{-1}} = \quad \therefore \theta = \begin{bmatrix} -76.25 \\ 7.5 \\ 7.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$\therefore$ Machine Model will be.

$$Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$Y = -76.25 + 7.5x_1 + 7.5x_2 + 0.5x_3$$

here, $x_1 \rightarrow$ Windspeed
$x_2 \rightarrow$ Blade angle.
$x_3 \rightarrow$ Rotor speed.

When,
$x_1 = 9$
$x_2 = 11$
$x_3 = 100$

$$Y = -76.25 + 7.5 \times 9 + 7.5 \times 11 + 0.5 \times 100$$

$$Y = 123.75 \text{ kW}$$